

Real Life Applications of System of Linear Equations

Example 1

The present age of Aditya and Mihir are in the ratio 2 : 1, and Mihir and Aymann are in the ratio 2 : 3. Fifteen years hence, their (Aditya and Mihir) ratio will be 7 : 5. **Formulate** the system of linear equations from this scenario and **Solve** the system of linear equations to find their present ages by using reduced row echelon method (RREM).

Solution:

Let us consider that the present ages of Aditya, Mihir and Aymann are x , y and z respectively.

Now according to the scenario,

1st condition: Ratio of the present ages of Aditya and Mihir is 2 : 1.

i.e.,

$$\frac{x}{y} = \frac{2}{1} \Rightarrow x = 2y \Rightarrow x - 2y = 0 \dots \dots \dots (1)$$

2nd condition: Ratio of the present ages of Mihir and Aymann is 2 : 3.

i.e.,

$$\frac{y}{z} = \frac{2}{3} \Rightarrow 3y = 2z \Rightarrow 3y - 2z = 0 \dots \dots \dots (2)$$

3rd condition: Fifteen years hence, ratio of the ages of Aditya and Mihir will be 7 : 5.

i.e.,

$$\frac{x+15}{y+15} = \frac{7}{5} \Rightarrow 5x + 75 = 7y + 105 \Rightarrow 5x - 7y = 105 - 75 \Rightarrow 5x - 7y = 30 \dots \dots \dots (3)$$

Now, we can write the following system of linear equations;

$$\begin{aligned} x - 2y &= 0 \\ 3y - 2z &= 0 \dots \dots \dots (4) \\ 5x - 7y &= 30 \end{aligned}$$

Now by applying reduced row echelon method we can get the value of x, y and z.

The augmented matrix is

$$\begin{bmatrix} 1 & -2 & 0 & 0 \\ 0 & 3 & -2 & 0 \\ 5 & -7 & 0 & 30 \end{bmatrix} \dots \dots \dots (5)$$

Now by using row operation:

$$\begin{bmatrix} 1 & -2 & 0 & 0 \\ 0 & 3 & -2 & 0 \\ 5 & -7 & 0 & 30 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 0 \\ 0 & 3 & -2 & 0 \\ 0 & 3 & 0 & 30 \end{bmatrix} \quad r'_3 = r_3 - 5r_1$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 0 \\ 0 & 1 & \frac{-2}{3} & 0 \\ 0 & 3 & 0 & 30 \end{bmatrix} \quad r'_2 = \frac{1}{3}r_2$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 0 \\ 0 & 1 & \frac{-2}{3} & 0 \\ 0 & 0 & 2 & 30 \end{bmatrix} \quad r'_3 = r_3 - 3r_2$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 0 \\ 0 & 1 & \frac{-2}{3} & 0 \\ 0 & 0 & 1 & 15 \end{bmatrix} \quad r'_3 = \frac{1}{2}r_3$$

Which is the row echelon form (REF).

Now for reduced row echelon,

$$\sim \begin{bmatrix} 1 & -2 & 0 & 0 \\ 0 & 1 & \frac{-2}{3} & 0 \\ 0 & 0 & 1 & 15 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 0 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & 15 \end{bmatrix} \quad r'_2 = r_2 + \frac{2}{3}r_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 20 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & 15 \end{bmatrix} \quad r'_1 = r_1 + 2r_2$$

Which is the reduced row echelon form (RREF).

Now by back substitution, we get $x = 20$, $y = 10$ and $z = 15$.

Hence, the present ages of Aditya, Mihir and Aymann are 20, 10 and 15 respectively.

Example 2

7 cups of coffee, 3 pieces of biscuits and 4 pieces of toast cost ₹500. 5 cups of coffee, 2 pieces of biscuits and 3 pieces of toast cost ₹360. 3 cups of coffee, 5 pieces of biscuits and 2 pieces of toast cost ₹260. **Formulate** the system of linear equations from this scenario and **Solve** the system of linear equations to find the cost of each item by using reduced row echelon method (RREM).

Solution:

Let us consider that the following prices: x for each cup of coffee, y for each biscuit and z for each piece of toast.

Now according to the scenario we can write the following system of linear equations;

$$\begin{aligned} 7x + 3y + 4z &= 500 \\ 5x + 2y + 3z &= 360 \dots \dots \dots (1) \\ 3x + 5y + 2z &= 260 \end{aligned}$$

Now by applying reduced row echelon method we can get the value of x , y and z .

(Try Yourself)

Assignment

1. The ratio between a two-digit number and the number obtained on reversing its digits is 7 : 4. If the difference between the digits of the numbers is 2. **Formulate** the system of linear equations from this scenario and **Solve** the system of linear equations to find the numbers by using reduced row echelon method (RREM).
2. The present age of Aditya and Mihir are in the ratio 2 : 1, and Mihir and Aymann are in the ratio 2 : 3. Fifth years hence, sum of their age will be 60 years. **Formulate** the system of linear equation from this scenario and **Solve** the system of linear equations to find their present ages by using reduced row echelon method (RREM).